

Model Merging for Medical LVLMs: A Benchmark and a Winner-Take-All Approach

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<https://github.com/MedAI-T/MergeMedBench>

Abstract. Large vision-language models (LVLMs) can be adapted to specialized medical imaging tasks via parameter-efficient fine-tuning approaches such as low-rank adaptation (LoRA), leading to a growing ecosystem of expert models tailored to specific imaging modalities and clinical scenarios. However, deploying multiple expert LVLMs in practice incurs substantial computational and operational overhead. Model merging provides a promising solution by consolidating multiple experts into a single model without retraining, yet it remains largely unexplored in the medical domain. In this work, we present the first systematic study of model merging for medical LVLMs. We introduce **MergeMed-Bench**, a comprehensive benchmark spanning eight imaging modalities and diverse clinical task types, comprising 16 LoRA fine-tuned models built upon two mainstream architectures. We conduct an extensive evaluation of existing merging methods and further propose winner-take-all, a simple and hyperparameter-free approach that retains only the most dominant parameters across expert models. By preserving the critical parameters that govern model behavior and discarding weaker ones, our method avoids the information dilution inherent in averaging- or alignment-based strategies. Despite its simplicity, winner-take-all consistently outperforms existing approaches, offering both a new perspective on LoRA merging and a strong practical baseline for future research.

Keywords: Model merging · Medical imaging · LVLMs · LoRA.

1 Introduction

Large vision-language models (LVLMs), such as Qwen-VL [1] and InternVL [3], have become foundational architectures for multimodal understanding. Beyond strong performance on general-purpose tasks, these models can be efficiently

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adapted to specialized domains through parameter-efficient fine-tuning, most notably low-rank adaptation (LoRA) [5]. This has enabled the development of domain-specific LVLMs, particularly in medical imaging, where fine-tuned models now exist for diverse imaging modalities and clinical scenarios.

However, deploying multiple specialized medical LVLMs introduces substantial computational and operational overhead. This raises a practical question: can we consolidate multiple expert medical LVLMs into a single model that performs well across diverse imaging modalities and clinical tasks? Model merging offers a promising solution by integrating parameters from independently fine-tuned models without retraining.

Various model merging methods have been proposed in natural language processing and computer vision [2,4,9,7,12,15,17,18,21,24,25,26]. [8] introduces task vectors—weight differences between fine-tuned and pre-trained models—and merges them via averaging. However, naive averaging can cause destructive interference when parameter updates conflict across tasks. TIES-Merging [20] addresses this by trimming redundant parameters and resolving sign conflicts before merging, while DARE [22] reduces redundancy by randomly dropping and rescaling delta parameters. More recently, methods targeting LoRA have emerged: KnOTS [19] observes that LoRA fine-tuned models exhibit lower weight alignment than fully fine-tuned counterparts and uses singular value decomposition (SVD) to transform LoRA weights into an aligned space; Core Space Merging [14] constructs a common subspace to improve merging; and Robust-Merge [23] emphasizes directional robustness when combining LoRA modules. Despite these advances, LoRA merging has not been systematically studied for medical LVLMs. Existing methods are developed and evaluated primarily on general language or vision benchmarks, leaving their effectiveness in the medical domain unexplored.

To address this gap, we present the first systematic study of model merging for medical LVLMs. We curate a comprehensive dataset spanning eight imaging modalities across multiple clinical task types. Using this dataset, we fine-tune 16 expert medical LVLMs with LoRA on two widely adopted architectures, Qwen-VL and InternVL, providing a rigorous testbed for evaluating model merging in medical settings.

We also propose a simple yet effective merging method called winner-take-all. Unlike existing approaches that rely on weighted averaging, sign consensus, or subspace alignment, our method follows a different principle: for each position in the LoRA matrices, we select and retain the most dominant parameter across all expert LVLMs. This strategy is motivated by the observation that a small subset of parameters disproportionately governs model behavior and is highly sensitive to modification. Preserving these dominant parameters exactly is crucial for maintaining task performance. In contrast, averaging- and alignment-based methods inevitably modify these critical parameters, causing information loss and performance degradation. Our contributions are:

- We present the first systematic study of model merging for medical LVLMs.

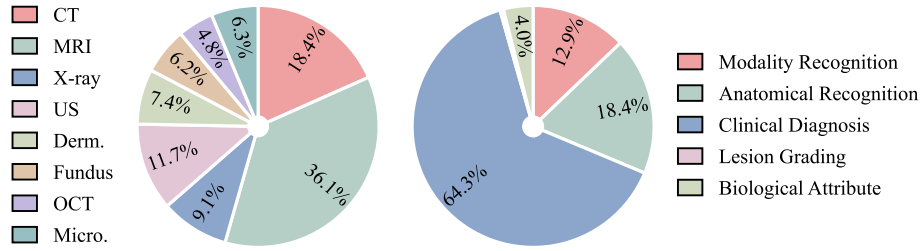


Fig. 1. Modality (left) and task (right) distributions of the evaluation set.

- We introduce MergeMedBench, a comprehensive benchmark spanning eight imaging modalities and multiple clinical task types, with 16 LoRA fine-tuned models on two mainstream LVLM architectures, to facilitate future research on medical model merging.
- We extensively benchmark existing model merging approaches on the proposed benchmark.
- We propose a hyperparameter-free LoRA merging method, winner-take-all, which consistently outperforms existing approaches despite its simplicity, offering both a new perspective on LoRA merging and a practical baseline for future research.

2 MergeMedBench

2.1 Dataset

We construct our benchmark using the open-access subset of the OmniMedVQA dataset [6], which contains 88,995 image-question-answer triplets spanning eight medical imaging modalities: CT, MRI, X-ray, ultrasound, dermoscopy, fundus photography, OCT, and microscopy. The dataset covers a wide range of clinically relevant tasks, including modality and anatomical recognition, clinical diagnosis, lesion grading, and biological attribute analysis.

We adopt a fixed training/test split to ensure reproducible evaluation and fair comparison. Specifically, 71,333 samples are used to fine-tune pre-trained LVLMs, while the remaining 17,662 samples are reserved for evaluation. Fig. 1 show the modality and task distributions of the evaluation set, respectively. The broad coverage of imaging modalities and task types enables comprehensive assessment of model merging performance in medical vision-language scenarios.

2.2 Medical LVLMs

We consider two widely adopted LVLM architectures: Qwen-VL and InternVL. Specifically, we use Qwen3-VL-2B-Instruct and InternVL2-1B as representative

models due to their strong performance and widespread adoption in vision-language research, particularly in the medical domain [10,13].

For each architecture, we fine-tune the pre-trained model on individual imaging modalities using LoRA. Following standard practice, LoRA modules are inserted into both attention and feed-forward layers, while base model parameters remain frozen. This yields a collection of expert LVLMs, each optimized for a particular imaging domain and clinical scenario. These expert models serve as inputs to the merging methods evaluated in our benchmark.

3 Methodology

3.1 On the Importance of Dominant Parameters

Before introducing our method, we present a key observation that motivates our approach. Within learned LoRA parameters, a small subset disproportionately governs model behavior; we call these dominant parameters. These parameters encode critical task-relevant information but are highly sensitive to modification. In contrast, non-dominant parameters can be altered substantially with little effect on performance. Consequently, identifying and preserving dominant parameters is essential for effective model merging.

To study this empirically, we conduct a controlled rescaling analysis on LoRA fine-tuned LVLMs. We partition LoRA parameters into dominant and non-dominant subsets based on parameter magnitude¹. We then independently rescale each subset by a factor $\alpha \in \{0, 0.1, \dots, 1.0, \dots, 3.0\}$ and measure the resulting performance. As shown in Fig. 2, rescaling dominant parameters causes dramatic performance shifts—both amplification ($\alpha > 1$) and attenuation ($\alpha < 1$) severely degrade accuracy. In contrast, rescaling non-dominant parameters over the same range produces virtually no change.

This observation has a critical implication for model merging: dominant parameters should be preserved as faithfully as possible. Existing merging methods

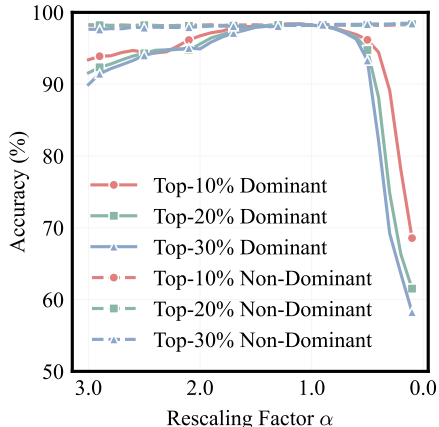


Fig. 2. Effect of rescaling dominant vs. non-dominant parameters. Dominant parameters are identified using a simple magnitude-based criterion for illustration. This motivates our winner-take-all merging strategy that prioritizes preserving dominant parameters.

¹ Magnitude provides a natural proxy for parameter importance; we discuss more general definitions of dominance later.

fail to satisfy this requirement. While approaches such as TIES-Merging [20] explicitly retain high-magnitude parameters, they ultimately rely on averaging to combine contributions from multiple models. This averaging inevitably modifies dominant parameters, causing information loss. The same limitation applies to subspace-based methods [14,19], which operate in transformed spaces but still employ averaging as the final aggregation step.

3.2 Problem Formulation

Consider a pre-trained LVLM with parameters θ_0 . Given N downstream tasks, we fine-tune θ_0 on each task independently using LoRA. For each dense layer l and weight matrix $M \in \{W_q, W_k, W_v, W_o\}$, LoRA introduces a low-rank update $\Delta W_{l,M} = B_{l,M} A_{l,M}$, where $B_{l,M} \in \mathbb{R}^{d \times r}$, $A_{l,M} \in \mathbb{R}^{r \times k}$, and the rank $r \ll \min(d, k)$. Let $\{(A_{l,M}^{(n)}, B_{l,M}^{(n)})\}_{n=1}^N$ denote the low-rank matrices from fine-tuning on the N tasks. The goal of model merging is to produce a single merged model $\theta_0 + \Delta\theta_{\text{merged}}$ that performs well across all N tasks without additional training, where $\Delta\theta_{\text{merged}}$ denotes all merged weight updates $\{\Delta W_{l,M}^{\text{merged}}\}_{l,M}$.

3.3 Method

We propose winner-take-all, a simple yet effective merging strategy grounded in the insight that dominant parameters should remain unmodified. Unlike existing methods that aggregate parameters through weighted averaging, sign consensus, or subspace projection, our approach follows a different principle: for each parameter position in the low-rank matrices, we select and retain the value from the expert that exhibits the strongest dominance according to a scoring function.

Scoring Function. We define a scoring function $\mathcal{S} : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ to quantify the dominance of individual parameter values. For each low-rank matrix $\Phi \in \{A, B\}$, at each position (i, j) in layer l and weight matrix M , the score for the n -th expert is:

$$\mathcal{S}_{l,M,i,j}^{(\Phi,n)} = \mathcal{S}(\Phi_{l,M}^{(n)}[i,j]) . \quad (1)$$

In principle, $\mathcal{S}$ can be instantiated in various ways. For example, gradient-based scoring such as $\mathcal{S}(w) = |w \cdot \nabla_w \mathcal{L}|$ incorporates task-specific sensitivity information but requires access to task data and incurs additional computation. In this work, we focus on gradient-free merging and use magnitude-based scoring.

A natural choice is $\mathcal{S}(w) = |w|$, reflecting the intuition that larger-magnitude parameters exert stronger influence on model activations. However, directly comparing raw magnitudes across different LoRA fine-tuned LVLMs can be suboptimal. LoRA parameter distributions may differ across experts due to differences in optimization trajectories and training data. Naive magnitude comparison may therefore select parameters that are large due to scale differences rather than true dominance, systematically favoring certain experts.

Table 1. Comparison of model merging methods based on Qwen-VL. We report accuracy (%) across eight imaging modalities (CT, MRI, X-ray, ultrasound, dermoscopy, fundus photography, OCT, and microscopy), along with average accuracy and runtime.

	CT	MRI	X-ray	US	Derm.	Fund.	OCT	Micro.	Avg.	Time
Individual	98.27	99.63	95.85	99.95	94.86	97.72	98.11	98.64	97.88	-
TA [8]	56.40	69.51	82.54	79.51	77.72	87.52	68.16	79.28	75.08	0.05
TIES-Merging [20]	86.45	90.08	91.21	88.28	83.00	91.71	79.95	84.59	86.91	1.20
DARE-TIES [22]	87.29	92.03	85.39	96.05	81.93	92.62	78.89	82.25	87.06	32.72
Iso-C [11]	52.48	63.96	81.30	75.36	75.19	82.33	66.16	76.04	71.60	13.52
Iso-CTS [11]	52.36	64.02	81.24	75.70	75.11	81.97	65.80	75.68	71.48	170.57
STF [16]	89.66	93.28	89.91	95.32	<u>84.84</u>	<u>93.53</u>	<u>87.50</u>	84.41	<u>89.81</u>	64.98
RobustMerge [23]	<u>90.19</u>	96.39	89.78	<u>95.47</u>	82.08	92.71	87.26	82.34	89.53	0.54
KnOTS [19]	63.59	79.70	86.69	84.28	80.25	90.07	76.06	81.44	80.26	272.46
KnOTS-TIES [19]	87.32	91.95	<u>91.33</u>	89.10	84.15	92.90	85.38	<u>85.23</u>	88.42	272.90
KnOTS-DARE-TIES [19]	89.23	93.80	88.54	92.67	83.92	92.71	92.22	84.41	89.69	303.85
Core [14]	56.31	69.48	82.72	79.51	78.10	87.25	67.92	79.73	75.13	650.57
Core-TIES [14]	65.04	79.72	86.69	86.31	79.79	90.35	78.30	80.90	80.89	644.03
Core-DARE-TIES [14]	76.92	80.58	76.41	80.28	73.28	83.33	71.11	74.86	77.10	635.65
Winner-Take-All	93.80	<u>96.30</u>	93.50	91.51	85.07	94.17	86.08	86.58	90.88	<u>0.21</u>

To address this, we introduce a normalized magnitude-based scoring scheme:

$$\mathcal{S}_{l,M,i,j}^{(\Phi,n)} = \left| \frac{\Phi_{l,M}^{(n)}[i,j] - \mathbb{E}[\Phi^{(n)}]}{\sqrt{\text{Var}[\Phi^{(n)}] + \epsilon}} \right|, \quad (2)$$

where $\mathbb{E}[\cdot]$ and $\text{Var}[\cdot]$ denote the mean and variance over all entries in $\Phi^{(n)}$, and ϵ is a small constant for numerical stability. Normalization ensures robust scoring across experts with different parameter scales.

Winner Selection. For each parameter position (i,j) at layer l and weight matrix M , we independently select the winning expert for each low-rank matrix $\Phi \in \{A, B\}$:

$$n_{l,M,i,j}^{*(\Phi)} = \arg \max_{n \in \{1, \dots, N\}} \mathcal{S}_{l,M,i,j}^{(\Phi,n)}. \quad (3)$$

The merged parameter at each position is then directly taken from the winning expert without rescaling or transformation:

$$\Phi_{l,M}^{\text{merged}}[i,j] = \Phi_{l,M}^{(n_{l,M,i,j}^{*(\Phi)})}[i,j]. \quad (4)$$

The complete merged LoRA update is:

$$\Delta W_{l,M}^{\text{merged}} = B_{l,M}^{\text{merged}} A_{l,M}^{\text{merged}}, \quad (5)$$

which is applied to the base model to obtain the final merged model $\theta_0 + \Delta\theta_{\text{merged}}$.

Table 2. Comparison of model merging methods based on InternVL.

	CT	MRI	X-ray	US	Derm.	Fund.	OCT	Micro.	Avg.	Time
Individual	99.81	99.73	98.58	99.95	98.09	99.73	100.00	99.55	99.43	-
TA [8]	63.68	62.75	77.21	45.23	58.27	80.15	65.92	74.32	65.94	0.03
TIES-Merging [20]	84.85	80.11	88.05	58.63	70.21	89.62	77.95	84.23	79.21	0.31
DARE-TIES [22]	78.99	84.46	79.20	48.55	<u>77.72</u>	81.88	75.35	69.91	74.51	4.56
Iso-C [11]	25.08	23.85	16.84	24.06	20.90	18.85	26.18	23.24	22.38	2.34
Iso-CTS [11]	25.58	24.80	16.78	25.22	23.28	18.76	28.18	23.96	23.32	38.21
STF [16]	89.42	85.75	<u>90.09</u>	<u>67.70</u>	75.65	93.26	74.76	86.67	<u>82.91</u>	15.65
RobustMerge [23]	81.33	<u>86.31</u>	88.36	51.69	77.11	88.34	79.13	78.56	78.85	0.26
KnOTS [19]	63.22	62.70	77.15	45.27	58.04	80.69	66.27	74.32	65.96	42.06
KnOTS-TIES [19]	82.84	82.28	87.55	67.60	68.53	88.89	87.50	<u>86.40</u>	81.45	42.51
KnOTS-DARE-TIES [19]	75.78	79.31	77.15	31.00	63.32	75.96	64.39	59.37	65.78	46.51
Core [14]	63.50	62.83	77.34	44.74	58.42	80.33	65.57	74.41	65.89	103.80
Core-TIES [14]	74.11	69.45	82.04	57.71	63.63	86.25	76.77	81.53	73.94	115.97
Core-DARE-TIES [14]	79.14	76.19	72.01	92.14	61.41	83.24	80.90	70.36	76.92	104.32
Winner-Take-All	<u>88.71</u>	89.97	91.02	62.15	81.09	<u>92.26</u>	<u>86.79</u>	85.14	84.64	<u>0.17</u>

4 Experiment

4.1 Baseline Methods and Metrics

We focus on gradient-free model merging methods for fair comparison with our winner-take-all approach. Specifically, we consider the following representative baselines: Task Arithmetic (TA) [8], TIES-Merging [20], DARE [22], Isotropic Merging [11], STF [16], RobustMerge [23], KnOTS [19], and Core Space Merging (Core) [14].

DARE is commonly combined with TIES-Merging, referred to as DARE-TIES. Isotropic Merging has two variants: Iso-C, which operates in a common subspace, and Iso-CTS, which operates in both common and task-specific subspaces. Some subspace-based methods, such as KnOTS and Core, integrate with TIES-Merging or DARE-TIES to further enhance merging performance. We denote these variants by appending the corresponding method names (e.g., KnOTS-TIES and Core-DARE-TIES).

We use normalized per-modality accuracy and average normalized accuracy across all eight modalities to evaluate merging performance. Following prior work, normalized accuracy is defined as the ratio of the merged model’s performance on each modality to that of the corresponding individually fine-tuned model. We also report merging time to assess computational efficiency.

4.2 Results

Tables 1 and 2 report the performance of all merging methods on our benchmark. Naive averaging-based TA performs poorly, while TIES-Merging improves results by trimming redundant parameters and resolving sign conflicts before averaging. DARE-TIES uses random dropping instead of deterministic trimming, yielding less stable performance. Among baselines, STF—a subspace method from

Table 3. Comparison of winner selection strategies in our framework using Qwen-VL.

k	1	2	3	4	5	6	7	8
Avg.	90.88	75.76	72.48	71.46	71.07	71.05	70.98	70.94

Table 4. Results of combining winner-take-all with subspace-based methods.

	Core	Core-Ours	KnOTS	KnOTS-Ours
Avg.	75.13	89.68	80.26	87.95

Table 5. Results of different normalization methods in the scoring function.

	w/o	min-max	RMS	Ours
Avg.	82.89	71.82	84.59	84.64

natural language processing—achieves consistently strong performance, whereas other subspace-based methods (e.g., KnOTS-DARE-TIES, Core-TIES) show unstable behavior.

In comparison, our winner-take-all method achieves the best average accuracy: 90.88% on Qwen3-VL-2B-Instruct and 84.64% on InternVL2-1B, consistently outperforming all baselines. Regarding efficiency, subspace-based methods incur substantially higher computational cost, whereas winner-take-all is faster than most baselines while delivering superior performance.

4.3 Discussion

Winner Selection Strategy. Table 3 evaluates different winner selection strategies, where k denotes selecting the k -th ranked expert at each parameter position based on the dominance score. Performance peaks at $k = 1$ and decreases as k increases, confirming that preserving the most dominant parameters is crucial.

Winner-Take-All versus Averaging. We compare winner-take-all with simple parameter averaging. Winner-take-all clearly outperforms averaging (90.88% vs. 72.18% in average accuracy), supporting our claim that averaging perturbs sensitive dominant parameters, whereas our method preserves them intact.

Combining Winner-Take-All with Subspace Approaches. We integrate winner-take-all with Core and KnOTS and compare the results with their original versions. As shown in Table 4, the combined methods improve average accuracy by 14.55% (Core) and 7.69% (KnOTS) when merging Qwen-VL fine-tuned models, showing our insight generalizes beyond raw parameter space. Notably, Core with winner-take-all outperforms both Core-TIES and Core-DARE-TIES, while performance remains comparable for KnOTS. These results indicate that winner-take-all is effective both in parameter space and subspace-based merging.

Effect of Normalization in the Scoring Function. Table 5 studies the impact of normalization on InternVL2, where the models exhibit pronounced misalignment. Our normalized magnitude-based scoring improves performance by 1.75% over the unnormalized variant. We further compare alternative normalization schemes (min-max and RMS), finding our formulation performs best.

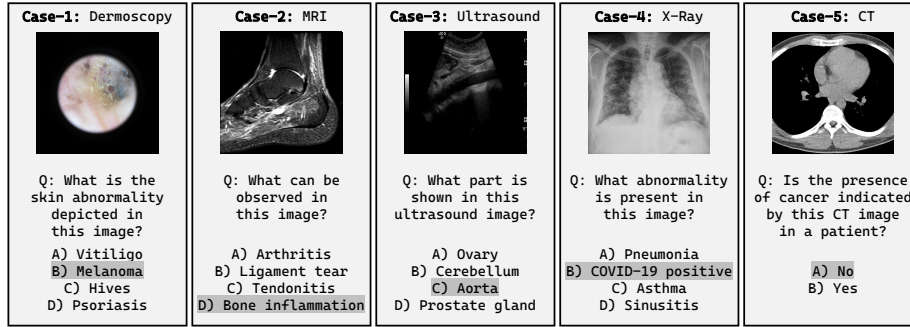


Fig. 3. Examples of our merged model on multiple modalities and tasks.

5 Conclusion

We present the first systematic study of model merging for medical LVLMs. Through extensive evaluation, we benchmark existing merging methods in the medical domain. Furthermore, we propose winner-take-all, a simple yet effective hyperparameter-free strategy that preserves dominant parameters during merging and consistently outperforms prior approaches. We hope this work establishes a strong foundation and spurs future research on efficient merging of medical LVLMs.

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